

Name: _____ Reg #: _____ Section: _____

National University of Computer and Emerging Sciences, Lahore Campus

Course: Artificial Intelligence

Program: BS(Computer Science)

Duration: 30 Minutes

Paper Date: 18-Feb-2026

Semester: Spring 2026

Course Code: AI2002

Section: BCS-6B

Total Marks: 15

Page(s): 2

Exam: Quiz 2

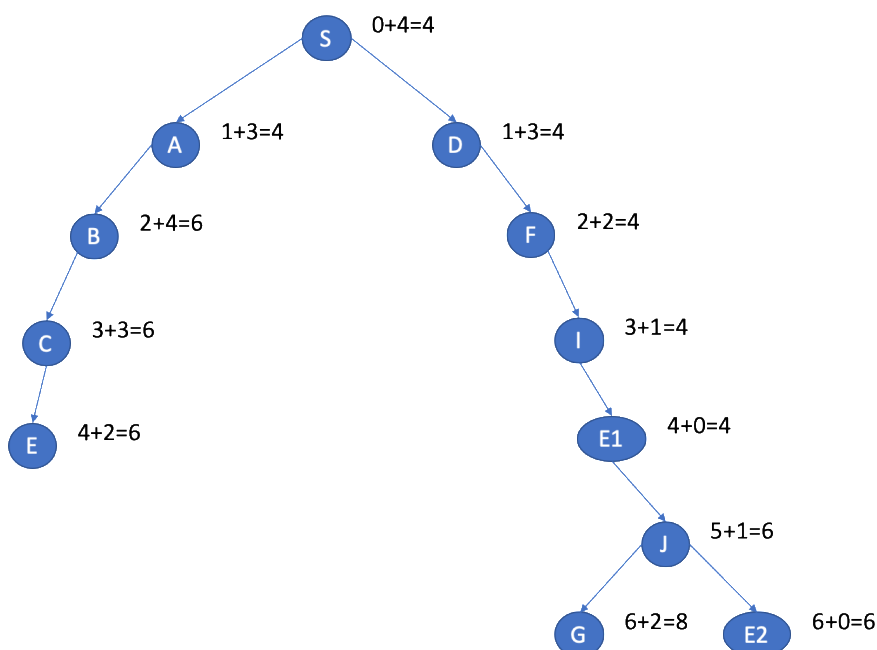
Q1) Inside a burning warehouse, a firefighter robot must guide workers to their nearest safe exits as fast as possible. Smoke blocks some paths.

S	A	B	C
D	X	X	E
F	X	G	H
I	<u>E1</u>	J	<u>E2</u>

- S = Robot start coordinates (1,1)
- E1 and E2 = Safe exits (Goals) at (4,2) and (4,4)
- X = Blocked by Smoke
- Move cost = 1
- Moves allowed = up, down, left, right
- The robot uses A* search with Manhattan Distance to the nearest goal as a heuristic.

$$h(n) = |x_1 - x_2| + |y_1 - y_2|$$

- (a) Draw the search tree Using A*. Calculate and write down the evaluation cost of each node. (4)



(b) Write the order in which nodes are expanded. (1)

S(4), A(4), D(4), F(4), I(4), E1(4), B(6), J(6), C(6)

(c) Must maintain both the frontier (open list) and the visited (closed list). (4)

Open / Frontier	Closed / Visited
s(4)	-
A(4),D(4)	S(4)
D(4), B(6)	A(4), S(4)
F(4) , B(6)	D(4), A(4), S(4)
I(4), B(6)	F(4), D(4), A(4), S(4)
E1(4), B(6)	I(4), F(4), D(4), A(4), S(4)
B(6), J(6)	E1(4), I(4), F(4), D(4), A(4), S(4)
J(6), C(6)	B(6), E1(4), I(4), F(4), D(4), A(4), S(4)
C(6), E2(6), G(8)	J(6), B(6), E1(4), I(4), F(4), D(4), A(4), S(4)
E2(6), G(6), E(6)	C(6), J(6), B(6), E1(4), I(4), F(4), D(4), A(4), S(4)

(d) Find the final solution path in which both exits are explored exactly once. (1)

S(4), D(4), F(4), I(4), E1(4), J(6), E2(6)

(e) Check whether the heuristic used are admissible. (4)

n	H(n)	Actual Cost	Admissible?
S	4	4	Yes
A	3	5	Yes
B	4	4	Yes
C	3	3	Yes
D	3	3	Yes
E	2	2	Yes
F	2	2	Yes
G	2	2	Yes
H	1	1	Yes
I	1	1	Yes
E1	0	0	Yes
J	1	1	Yes
E2	0	0	Yes

(f) When the solution is found, mention the states that remain in the frontier. (1)

E2(6), G(6), E(6)